$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathcal{S}}(t) = -\mathrm{i}\big[\hat{H}_{\mathcal{S}}, \hat{\rho}_{\mathcal{S}}(t)\big] - \int_0^\infty \mathrm{d}\tau\, \{\nu(\tau)\,[\hat{\sigma}_z, [\hat{\sigma}_z(-\tau), \hat{\rho}_{\mathcal{S}}(t)]] \\ -\mathrm{i}\eta(\tau)\,[\hat{\sigma}_z, \{\hat{\sigma}_z(-\tau), \hat{\rho}_{\mathcal{S}}(t)\}]\}\,, \quad (5.138)$$